

MADSON ALLAN DE LUNA ARAGÃO

Data Scientist | Data Engineer | Full-Stack Developer

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madsondeluna.com | GitHub | LinkedIn | Google Scholar | ORCID

PROFESSIONAL SUMMARY

Data, software engineering and digital product professional with more than six years of experience in software development and data analysis since 2020. Works as a full-stack developer, combining front-end experience with React, TypeScript, JavaScript, HTML, CSS, Vite, Recharts and data-driven interfaces, and back-end experience with Python, FastAPI, Flask, REST and GraphQL APIs, Pydantic, SQLAlchemy, PostgreSQL, MySQL, SQLite, Redis and service integrations. Builds end-to-end solutions spanning data modeling and ETL pipelines through automated testing, Docker, documentation and Linux and HPC execution. Experienced with classical machine-learning models including regression, Random Forest, SVM, KNN, XGBoost and LightGBM, as well as deep learning, especially neural networks, Transformer architectures and large language models (LLMs). Develops and researches applications in this area, recognizing that LLMs are deep-learning models built primarily on Transformer architectures. Also experienced in technical leadership and management of software development teams, translating business needs into requirements, roadmaps and deliverables. Formal training includes Data Science and Analytics at the Pontifical Catholic University of Rio de Janeiro and an MBA in Software Engineering at the University of São Paulo, Brazilian institutions recognized nationally and across Latin America.

SKILLS

Programming and data: Python, SQL, JavaScript, TypeScript, Bash/Shell and R; working knowledge of Java; HTML and CSS; pandas, NumPy and SciPy.

Front end: React, TypeScript, JavaScript, HTML, CSS, Vite, Recharts, Plotly and data-driven web interfaces.

Back end: Python, FastAPI, Flask, Pydantic, SQLAlchemy, REST and GraphQL APIs, service integration, data validation and asynchronous applications.

Full stack and data engineering: End-to-end application development; ETL, data modeling and governance; PostgreSQL, MySQL, SQLite and Redis.

Machine learning: scikit-learn, XGBoost, LightGBM, linear and logistic regression, Random Forest, SVM, KNN, gradient boosting, feature engineering, PCA and hyperparameter tuning.

Deep learning, Transformers and LLMs: TensorFlow/Keras, CNNs and OpenCV in projects; Transformer architectures, generative models and LLMs in projects and applied research; cross-validation, F1, ROC-AUC, MAE, RMSE and R².

Quality and infrastructure: Git/GitHub, pytest, mock-based tests, Docker/Compose, Linux/Unix, Slurm/HPC, Nextflow, Snakemake, CLI development, documentation and version control.

Technical leadership and products: Management and coordination of technical software development teams; Product Management and Product Ownership; requirements, backlog, MVPs, roadmaps, KPIs, Scrum, Kanban, Lean and user research.

Communication and tools: Stakeholder management, cross-functional collaboration, active listening, critical thinking, problem-solving and data storytelling; Jira, Trello, Figma, Miro and Notion.

PROFESSIONAL EXPERIENCE

Biotech Business

Oct 2025 - Present

- Head of Data Science since Aug 2026; previous responsibilities in software development and data analysis from Oct 2025 to Jul 2026.
- Develop machine learning and computer vision applications, APIs and database integrations to support data analysis and research processes.
- Translate business needs into requirements, priorities and deliverables; contribute to product roadmaps and alignment with technical and business stakeholders.

PickCells | Research and Development Analyst

Oct 2020 - Aug 2023 | Recife, Brazil

- Prototyped, developed and validated AI, computer vision and IoT products; defined requirements, built MVPs and contributed to roadmaps and KPI tracking.
- Conducted market and user research, collected feedback and prioritized improvements throughout the product lifecycle; collaborated with technical teams and stakeholders using Agile, Lean and Scrum.

Applied research: UFMG (Aug 2024 - Present), working with generative models and computational workflows; UFPE (Jun 2022 - Jul 2024), covering data preparation, feature engineering and AMPIdentifier development. Earlier R&D internship at PickCells (Apr - Aug 2020).

DATA AND SOFTWARE PROJECTS

AMPIdentifier | Classification models and Python package

Python, scikit-learn, XGBoost, LightGBM, joblib

Feature extraction, classifier training and comparison, hyperparameter search, and inference through CLI and web interfaces. Distributed as a PyPI package.

GenVar | Data integration and exploration platform

Python, FastAPI, JavaScript/React, Redis, Docker, pytest

Integrates external sources through a web application with an asynchronous API, response validation, caching and visualizations. Includes mock-based service tests and error handling.

Professional portfolio | Web development

HTML, CSS, JavaScript, TypeScript

Developed a portfolio website showcasing projects, applications and technical work, bringing interfaces and content together on a dedicated site.

Open source | Seqera, Nextflow and the nf-core ecosystem

Python, graphs, testing and documentation

Nextflow Ambassador. Merged contribution to Seqera's nf-metro: diagnostics for connections removed during graph conversion, with tests and documentation.

eIF4E Atlas | ETL pipeline and data warehouse

Python, SQL, SQLite, APIs, JavaScript

API data collection, transformation and persistence in a dimensional model, SQL queries and data exports for web-based exploration.

Skills for LLM agents | Developer tooling

Python, TypeScript, Rich, Ink, CLI/TUI

Created reusable skills to guide LLM-based agents in terminal interface development, combining structured instructions, templates and verification scripts.

More projects, libraries and applications available at github.com/madsondeluna

COMMUNITY AND OPEN-SOURCE CONTRIBUTIONS

Nextflow Ambassador | Nextflow / Seqera

Jan 2026 - Present

- Support researchers and teams in adopting Nextflow and migrating workflows, applying reproducibility, automation and execution practices across Linux and HPC environments.
- Share technical knowledge, guide users and collaborate within the Nextflow and nf-core communities, connecting practical needs with appropriate tooling.
- Contribute code, tests and documentation to open-source tools in the Seqera ecosystem, including an improvement merged into nf-metro.

OUTPUTS AND INTELLECTUAL PROPERTY

2 software registrations, 2 patent applications related to the development of diagnostic techniques, and 4 software registration applications under review at INPI - Instituto Nacional da Propriedade Industrial (Brazilian National Institute of Industrial Property). Registrations include AMPIdentifier, BR 51 2025 005859-4. Scientific output: six journal articles and one chapter, referenced on Google Scholar and ORCID.

LANGUAGES

Portuguese: native. English: advanced.

EDUCATION

MBA in Software Engineering: University of São Paulo (USP) | In progress; expected completion December 2026

Postgraduate Specialization in Data Science and Analytics: PUC-Rio | 2026

Master's degree: Federal University of Pernambuco (UFPE) | 2024

Bachelor's degree: Federal University of Pernambuco (UFPE) | 2022

Technical Diploma in Software Development: Escola Técnica Estadual de Pernambuco | 2013

CERTIFICATIONS AND COURSES

Aug 2026 | Data Responsibility - Google

Aug 2026 | Stakeholder Management - Google

Aug 2026 | Make Data-Driven Decisions - Google

Aug 2026 | Data Analytics and Analytical Thinking - Google

Jun 2026 | Docker Foundations Professional Certificate - Docker, Inc.

Dec 2025 | Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate Certification - Oracle

Oct 2025 | Machine Learning and Analytics - CCEC / PUC-Rio

Sep 2025 | Nextflow Training - Seqera

Sep 2025 | Notion Workflows - Notion | Valid through Sep 2027

May 2025 | Agile Project Management Professional Certificate - Atlassian

Apr 2025 | Generative AI - MIT Professional Education

Apr 2025 | Requirements Engineering and Agile Product Management - PUC-Rio

Mar 2025 | Microsoft Azure AI Essentials: Workloads and Machine Learning - Microsoft

Jun 2024 | Advanced Gemini for Developers

Jan 2024 | Career Essentials in GitHub Professional Certificate - GitHub

Aug 2022 | Multidimensional Arrays and Their Traversal in Python - CodeSignal

Jun 2022 | Slurm Workload Manager - Laboratório Nacional de Computação Científica